

```
big@hell-na:~/Kishor$ curl google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html; charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="http://www.google.com/">here</A>.
</BODY></HTML>
big@hell-na:~/Kishor$ curl kishorneupane.com.np
<html>
<head><title>301 Moved Permanently</title></head>
<body>
<center><h1>301 Moved Permanently</h1></center>
<hr><center>cloudflare</center>
</body>
</html>
```

```
Jun 25 18:20
big@hell-na: ~/Kishor
big@hell-na:~/Kishor$ nc -vz localhost 80
Connection to localhost (127.0.0.1) 80 port [tcp/http] succeeded!
big@hell-na:~/Kishor$
```

```
big@hell-na:~/Kishor$ nmap kishorneupane.com.np
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-06-25 18:09 +0545
Nmap scan report for kishorneupane.com.np (172.67.217.170)
Host is up (0.049s latency).
Other addresses for kishorneupane.com.np (not scanned): 2606:4700:3030::ac43:d9aa 2606:4700:3032::6815:3e0b 104.21.62.11
Not shown: 995 filtered tcp ports (no-response), 1 filtered tcp ports (host-unreach)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https
8080/tcp   open  http-proxy
8443/tcp   open  https-alt

Nmap done: 1 IP address (1 host up) scanned in 5.66 seconds
```

```
big@hell-na:~/Kishor$ ip route show
default via 192.168.1.254 dev wlo1 proto dhcp src 192.168.1.71 metric 600
10.0.3.0/24 dev lxcbr0 proto kernel scope link src 10.0.3.1 linkdown
192.168.1.0/24 dev wlo1 proto kernel scope link src 192.168.1.71 metric 600
big@hell-na:~/Kishor$ route -n
Kernel IP routing table
Destination      Gateway         Genmask        Flags Metric Ref    Use Iface
0.0.0.0          192.168.1.254  0.0.0.0        UG      600    0      0 wlo1
10.0.3.0         0.0.0.0        255.255.255.0  U       0      0      0 lxcbr0
192.168.1.0      0.0.0.0        255.255.255.0  U       600    0      0 wlo1
big@hell-na:~/Kishor$
```



```
big@hell-na:~/Kishor$ nslookup kishorneupane.com.np
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   kishorneupane.com.np
Address: 104.21.62.11
Name:   kishorneupane.com.np
Address: 172.67.217.170
Name:   kishorneupane.com.np
Address: 2606:4700:3032::6815:3e0b
Name:   kishorneupane.com.np
Address: 2606:4700:3030::ac43:d9aa
```

```
root@hell-na: /home/big/Kishor
root@hell-na:/home/big/Kishor# netstat -tulnp
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN      758/systemd-resolve
tcp        0      0 10.0.3.1:53             0.0.0.0:*               LISTEN      2094/dnsmasq
tcp        0      0 127.0.0.1:8081           0.0.0.0:*               LISTEN      4977/bettercap
tcp        0      0 127.0.0.1:631            0.0.0.0:*               LISTEN      1381/cupsd
tcp        0      0 127.0.0.54:53            0.0.0.0:*               LISTEN      758/systemd-resolve
tcp        0      0 127.0.0.1:9050           0.0.0.0:*               LISTEN      2070/tor
tcp6       0      0 :::80                    :::*                    LISTEN      1426/apache2
tcp6       0      0 :::1:631                  :::*                    LISTEN      1381/cupsd
udp        0      0 0.0.0.0:40003            0.0.0.0:*               1065/avahi-daemon:
udp        0      0 0.0.0.0:5353             0.0.0.0:*               1065/avahi-daemon:
udp        0      0 10.0.3.1:53              0.0.0.0:*               2094/dnsmasq
udp        0      0 127.0.0.54:53            0.0.0.0:*               758/systemd-resolve
udp        0      0 127.0.0.53:53            0.0.0.0:*               758/systemd-resolve
udp        0      0 0.0.0.0:67               0.0.0.0:*               2094/dnsmasq
udp6       0      0 :::5353                  :::*                    1065/avahi-daemon:
udp6       0      0 :::32864                  :::*                    1065/avahi-daemon:
udp6       0      0 fe80::2c49:90f9:a79:546 :::*                    1108/NetworkManager
```

```

big@hell-na:~/Kishor$ traceroute kishorneupane.com.np
traceroute to kishorneupane.com.np (104.21.62.11), 30 hops max, 60 byte packets
 1 dsldevice.lan (192.168.1.254)  5.214 ms  5.223 ms  5.292 ms
 2 datahub-bng2.subisu.net.np (116.66.192.1)  5.655 ms  5.615 ms  5.577 ms
 3 116.66.193.95 (116.66.193.95)  5.541 ms  5.483 ms  5.456 ms
 4 116.66.193.11 (116.66.193.11)  9.860 ms  9.854 ms  9.813 ms
 5 * * *
 6 182.93.69.255 (182.93.69.255)  12.003 ms  7.918 ms  8.286 ms
 7 104.21.62.11 (104.21.62.11)  8.606 ms  8.610 ms  8.587 ms
big@hell-na:~/Kishor$ traceroute google.com
traceroute to google.com (142.250.134.101), 30 hops max, 60 byte packets
 1 dsldevice.lan (192.168.1.254)  5.170 ms  5.121 ms  5.087 ms
 2 datahub-bng2.subisu.net.np (116.66.192.1)  9.033 ms  9.002 ms  8.967 ms
 3 116.66.193.95 (116.66.193.95)  8.956 ms  8.910 ms  8.904 ms
 4 116.66.193.92 (116.66.193.92)  8.883 ms  8.860 ms  8.836 ms
 5 116.66.193.253 (116.66.193.253)  11.900 ms  11.895 ms  11.852 ms
 6 * * *
 7 * * *
 8 * * *
 9 142.251.52.204 (142.251.52.204)  26.398 ms  142.251.76.168 (142.251.76.168)  25.505 ms  142.251.49.120 (142.251.49.120)  26.249 ms
10 192.178.83.226 (192.178.83.226)  62.052 ms  216.239.62.218 (216.239.62.218)  24.528 ms  216.239.62.180 (216.239.62.180)  25.481 ms
11 192.178.124.243 (192.178.124.243)  46.208 ms  45.674 ms  48.943 ms
12 192.178.254.236 (192.178.254.236)  43.463 ms  45.722 ms  192.178.254.210 (192.178.254.210)  44.740 ms
13 142.251.78.223 (142.251.78.223)  43.888 ms  142.251.226.113 (142.251.226.113)  45.713 ms  172.253.65.15 (172.253.65.15)  45.848 ms
14 142.250.208.82 (142.250.208.82)  43.150 ms  142.250.236.126 (142.250.236.126)  44.232 ms  142.251.244.184 (142.251.244.184)  45.225 ms
15 142.250.236.126 (142.250.236.126)  44.181 ms  142.251.244.184 (142.251.244.184)  42.646 ms *
16 * * *
17 * * *
18 * * *
19 * * *
20 * * *
21 fx-in-f101.1e100.net (142.250.134.101)  46.906 ms  43.735 ms  43.651 ms

```

```
big@hell-na:~/Kishor$ hostname  
hell-na  
big@hell-na:~/Kishor$
```



```
big@hell-na:~/Kishor$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: wlo1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether a4:97:b1:d6:35:f1 brd ff:ff:ff:ff:ff:ff
    altname wlp1s0
    inet 192.168.1.71/24 brd 192.168.1.255 scope global dynamic noprefixroute wlo1
        valid_lft 82676sec preferred_lft 82676sec
    inet6 2403:3800:3240:eb4a:669b:cd1f:efe2:950a/64 scope global temporary dynamic
        valid_lft 86276sec preferred_lft 82118sec
    inet6 2403:3800:3240:eb4a:3083:dfc5:1965:e209/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86276sec preferred_lft 86276sec
    inet6 fe80::2c49:90f9:a793:7fa/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: lxcbr0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default qlen 1000
    link/ether 00:16:3e:00:00:00 brd ff:ff:ff:ff:ff:ff
    inet 10.0.3.1/24 brd 10.0.3.255 scope global lxcbr0
        valid_lft forever preferred_lft forever
```

```
Windows PowerShell
PS C:\Users\Kisu> getmac

Physical Address      Transport Name
=====
A4-97-B1-D6-35-F1     \Device\Tcpip_{E030F551-B11C-423D-B782-9A4DE40E1018}
A4-97-B1-D6-35-F2     Media disconnected
PS C:\Users\Kisu> |
```

```
big@hell-na:~/Kishor$ ping -4 kishorneupane.com.np
PING kishorneupane.com.np (172.67.217.170) 56(84) bytes of data.
64 bytes from 172.67.217.170: icmp_seq=1 ttl=56 time=56.2 ms
64 bytes from 172.67.217.170: icmp_seq=2 ttl=56 time=41.0 ms
64 bytes from 172.67.217.170: icmp_seq=3 ttl=56 time=42.8 ms
64 bytes from 172.67.217.170: icmp_seq=4 ttl=56 time=40.9 ms
64 bytes from 172.67.217.170: icmp_seq=5 ttl=56 time=42.5 ms
64 bytes from 172.67.217.170: icmp_seq=6 ttl=56 time=42.5 ms
64 bytes from 172.67.217.170: icmp_seq=7 ttl=56 time=41.4 ms
64 bytes from 172.67.217.170: icmp_seq=8 ttl=56 time=40.6 ms
64 bytes from 172.67.217.170: icmp_seq=9 ttl=56 time=41.4 ms
64 bytes from 172.67.217.170: icmp_seq=10 ttl=56 time=40.9 ms
64 bytes from 172.67.217.170: icmp_seq=11 ttl=56 time=41.4 ms
64 bytes from 172.67.217.170: icmp_seq=12 ttl=56 time=42.6 ms
64 bytes from 172.67.217.170: icmp_seq=13 ttl=56 time=40.6 ms
64 bytes from 172.67.217.170: icmp_seq=14 ttl=56 time=40.7 ms
64 bytes from 172.67.217.170: icmp_seq=15 ttl=56 time=42.2 ms
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  QUERY RESULTS

● big@hell-na:~/Kishor/os_lab$ python3 15.py
Enter number of processes: 3
Max need of P0: 7
Max need of P1: 5
Max need of P2: 3
Allocated to P0: 3
Allocated to P1: 2
Allocated to P2: 2
Available instances: 3
○ Safe Sequence: P1 P2 P0 big@hell-na:~/Kishor/os_lab$
```

Output:

```
Enter Available Resources: 3 3 2
• big@hell-na:~/Kishor/os_lab$ python3 14.py
Enter number of processes: 5
Enter number of resource types: 3
Enter Allocation Matrix
0 1 0
2 0 0
3 0 2
2 1 1
0 0 2
Enter Max Matrix
7 5 3
3 2 2
9 0 2
2 2 2
4 3 3
Enter Available Resources: 3 3 2
○ Safe Sequence: P1 P3 P4 P0 P2 big@hell-na:~/Kishor/os_lab$
```

⊗ 0 ⚠ 0

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    QUERY RESULTS

● big@hell-na:~/Kishor/os_lab$ python3 13.py
P1 entered critical section
P1 left critical section
P2 entered critical section
P2 left critical section
○ big@hell-na:~/Kishor/os_lab$
```